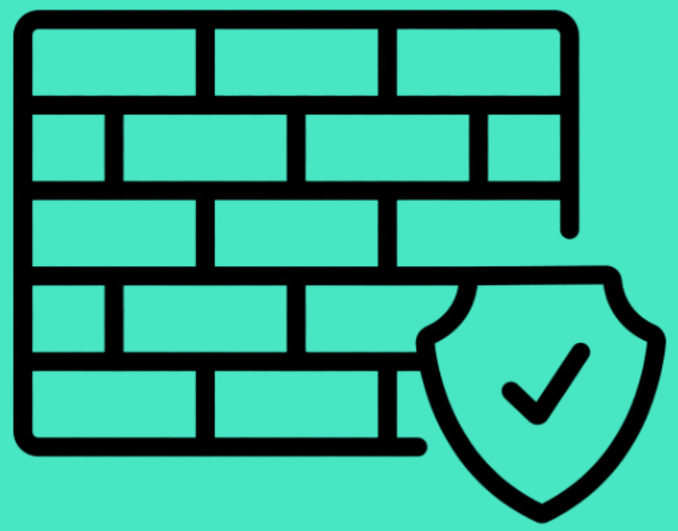
Firewall Team Project



Team Pizza

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팀원: 박찬우 이재호 김윤서

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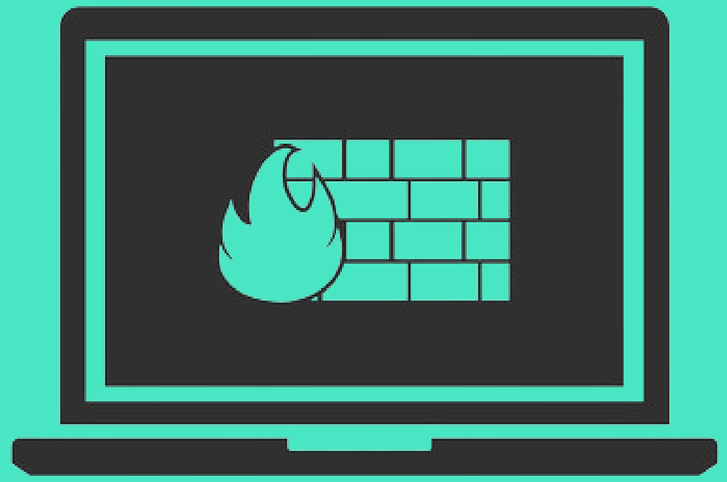
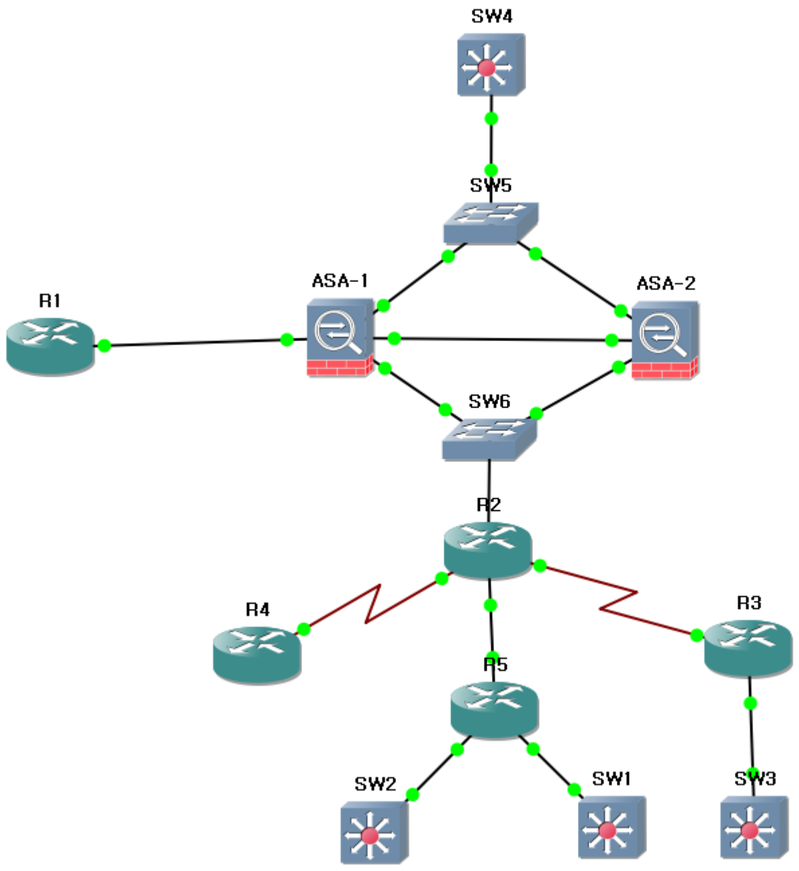


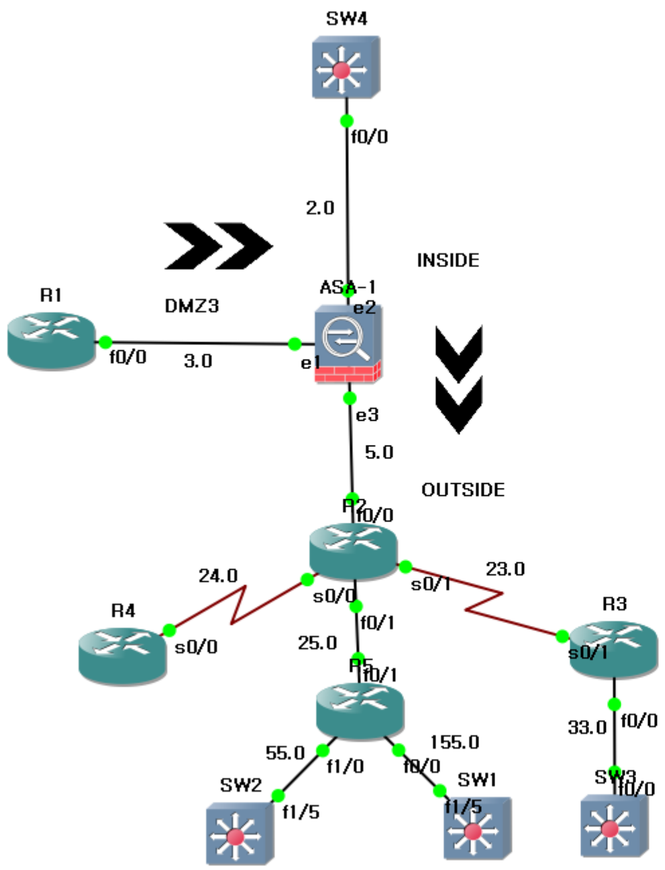
Chapter 1

물리적 / 논리적 구성도

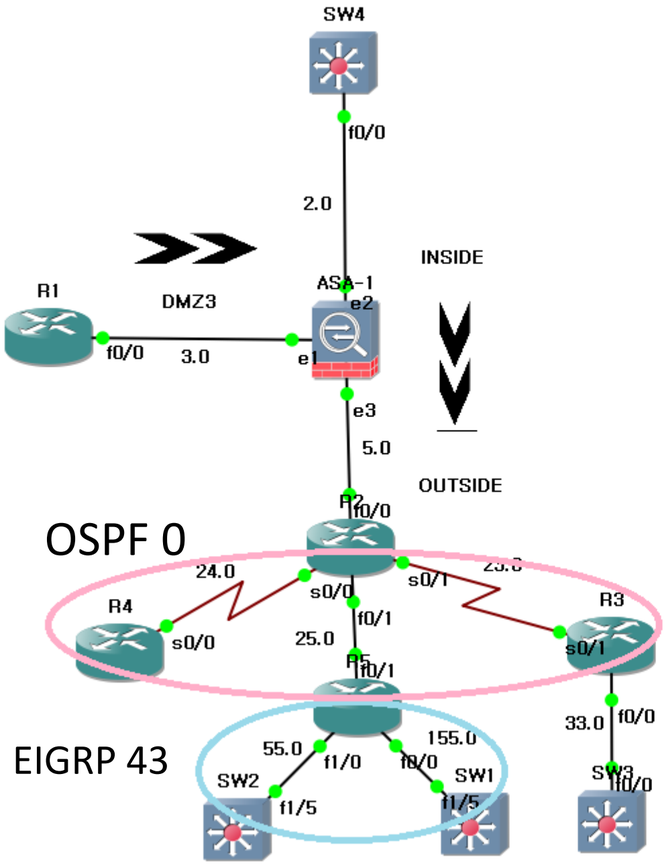
물리적 구성도



논리적 구성도


SW4
f0/0
2.0
INSIDE
ASA-1
e2
R1
DMZ3
f0/0
3.0
e1
e3
5.0
OUTSIDE
R2
f0/0
24.0
s0/0
s0/1
23.0
R4
s0/0
f0/1
25.0
R3
s0/1
R5
f0/1
33.0
f0/0
55.0
f1/0
155.0
f0/0
SW2
f1/5
SW1
f1/5
SW3
f0/0

논리적 구성도(영역)


SW4
f0/0
2.0
INSIDE
ASA-1
e2
R1
DMZ3
f0/0
3.0
e1
e3
5.0
OUTSIDE
R2
f0/0
OSPF 0
24.0
s0/0
s0/1
R4
s0/0
f0/1
25.0
R3
s0/1
R5
f0/1
33.0
f0/0
EIGRP 43
55.0
f1/0
155.0
f0/0
SW2
f1/5
SW1
f1/5
SW3
f0/0

Chapter 2

라우터 / 스위치 구성 및 주소 설정

라우터/스위치 구성 및 주소 설정

R1

```
int lo0
ip add 43.43.0.1
255.255.255.255

int lo100
ip add 111.111.111.111
255.255.255.0

int f0/0
no sh
ip add 43.43.3.1
255.255.255.0
```

R2

```
int lo0
ip add 43.43.0.2
255.255.255.255

int lo100
ip add 222.222.222.222
255.255.255.255

int f0/0
no sh
ip add 43.43.5.2
255.255.255.0

int f0/1
no sh
ip add 43.43.25.2 255.255.255.0

int s0/0
no sh
ip add 43.43.24.2 255.255.255.0
ip os net broadcast

int s0/1
no sh
ip add 43.43.23.2 255.255.255.0
ip os net broadcast
```

라우터/스위치 구성 및 주소 설정

R3

```
int lo0
ip add 43.43.0.3
255.255.255.255

int f0/0
no sh
ip add 43.43.33.3
255.255.255.0

int s0/1
no sh
ip add 43.43.23.3
255.255.255.0
ip os net broadcast
ip os priority 0
```

R4

```
int lo0
ip add 43.43.0.4 255.255.255.255

int s0/0
no sh
ip add 43.43.24.4 255.255.255.0
ip os net broadcast
ip os priority 0
int lo0
ip add 43.43.0.5 255.255.255.255
```

R5

```
int lo100
ip add 155.155.155.155
255.255.255.255

int f0/0
no sh
ip add 43.43.155.5 255.255.255.0

int f0/1
no sh
ip add 43.43.25.5 255.255.255.0

int f1/0
no sh
ip add 43.43.55.5 255.255.255.0
```

라우터/스위치 구성 및 주소 설정

SW1

```
int f1/5
no sh
no sw
ip add 43.43.155.250
255.255.255.0
```

SW2

```
int f1/5
no sh
no sw
ip add 43.43.55.250
255.255.255.0
```

SW3

```
int f1/5
no sh
no sw
ip add 43.43.155.250
255.255.255.0
```

SW4

```
int lo0
ip add 150.1.43.10
255.255.255.0

int f1/10
no sh
no sw
ip add 43.43.2.250
255.255.255.0
```

Chapter 2-(4) R1, R2, R3, R4 주소 설정 확인

라우터/스위치 구성 및 주소 설정-라우터 주소 설정 확인(1)

R1

```
R1(config-if)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        43.43.3.1       YES manual up                    up
Serial0/0              unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1              unassigned      YES unset  administratively down down
FastEthernet1/0        unassigned      YES unset  administratively down down
Loopback0              43.43.0.1       YES manual up                    up
Loopback100            111.111.111.111 YES manual up                    up
R1(config-if)#
```

R3

```
R3(config-if)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        43.43.33.3      YES manual up                    up
Serial0/0              unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1              43.43.23.3      YES manual up                    up
FastEthernet1/0        unassigned      YES unset  administratively down down
Loopback0              43.43.0.3       YES manual up                    up
R3(config-if)#
```

R2

```
R2(config-if)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        43.43.5.2       YES manual up                    up
Serial0/0              43.43.24.2      YES manual up                    up
FastEthernet0/1        43.43.25.2      YES manual up                    up
Serial0/1              43.43.23.2      YES manual up                    up
FastEthernet1/0        unassigned      YES unset  administratively down down
Loopback0              43.43.0.2       YES manual up                    up
Loopback100            222.222.222.222 YES manual up                    up
R2(config-if)#
```

R4

```
R4(config-if)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
Serial0/0              43.43.24.4      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1              unassigned      YES unset  administratively down down
FastEthernet1/0        unassigned      YES unset  administratively down down
Loopback0              43.43.0.4       YES manual up                    up
R4(config-if)#
```

라우터/스위치 구성 및 주소 설정-라우터 주소 설정 확인(2)

R5

```
R5(config-if)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        43.43.155.5     YES manual up                    up
Serial0/0              unassigned      YES unset  administratively down down
FastEthernet0/1        43.43.25.5      YES manual up                    up
Serial0/1              unassigned      YES unset  administratively down down
FastEthernet1/0        43.43.55.5      YES manual up                    up
Loopback0              43.43.0.5       YES manual up                    up
Loopback100            155.155.155.155 YES manual up                    up
R5(config-if)#
```

Chapter 2-(6) SW1, SW2, SW3, SW4 주소 설정 확인

라우터/스위치 구성 및 주소 설정-스위치 주소 설정 확인

SW1

```
SW1(config-if)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
Serial0/0              unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1              unassigned      YES unset  administratively down down
FastEthernet1/0        unassigned      YES unset  up                    down
FastEthernet1/1        unassigned      YES unset  up                    down
FastEthernet1/2        unassigned      YES unset  up                    down
FastEthernet1/3        unassigned      YES unset  up                    down
FastEthernet1/4        unassigned      YES unset  up                    down
FastEthernet1/5        43.43.155.250   YES manual up                    up
FastEthernet1/6        unassigned      YES unset  up                    down
 --More--
```

SW3

```
SW3(config-if)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
Serial0/0              unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1              unassigned      YES unset  administratively down down
FastEthernet1/0        unassigned      YES unset  up                    down
FastEthernet1/1        unassigned      YES unset  up                    down
FastEthernet1/2        unassigned      YES unset  up                    down
FastEthernet1/3        unassigned      YES unset  up                    down
FastEthernet1/4        unassigned      YES unset  up                    down
FastEthernet1/5        43.43.155.250   YES manual up                    down
FastEthernet1/6        unassigned      YES unset  up                    down
 --More--
```

SW2

```
SW2(config-if)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
Serial0/0              unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1              unassigned      YES unset  administratively down down
FastEthernet1/0        unassigned      YES unset  up                    down
FastEthernet1/1        unassigned      YES unset  up                    down
FastEthernet1/2        unassigned      YES unset  up                    down
FastEthernet1/3        unassigned      YES unset  up                    down
FastEthernet1/4        unassigned      YES unset  up                    down
FastEthernet1/5        43.43.55.250    YES manual up                    up
FastEthernet1/6        unassigned      YES unset  up                    down
 --More--
```

SW4

```
SW4(config-if)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
Serial0/0              unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/1              unassigned      YES unset  administratively down down
FastEthernet1/0        unassigned      YES unset  up                    down
FastEthernet1/1        unassigned      YES unset  up                    down
FastEthernet1/2        unassigned      YES unset  up                    down
FastEthernet1/3        unassigned      YES unset  up                    down
FastEthernet1/4        unassigned      YES unset  up                    down
FastEthernet1/5        unassigned      YES unset  up                    down
FastEthernet1/6        unassigned      YES unset  up                    down
FastEthernet1/7        unassigned      YES unset  up                    down
FastEthernet1/8        unassigned      YES unset  up                    down
FastEthernet1/9        unassigned      YES unset  up                    down
FastEthernet1/10       43.43.2.250     YES manual up                    up
```

Chapter 3

라우터 / 스위치
라우팅 설정

라우터/스위치 라우팅 설정 (1)

R1

```
ip route 0.0.0.0 0.0.0.0
43.43.3.253
```

R2

```
router os 1
router-id 43.43.0.2

net 43.43.24.2 0.0.0.0 a 0
net 43.43.23.2 0.0.0.0 a 0
net 43.43.25.2 0.0.0.0 a 0

ip route 0.0.0.0 0.0.0.0
43.43.5.253

ip route 43.43.3.0 255.255.255.0
43.43.5.253

default-information originate
```

R3

```
router os 1
router-id 43.43.0.3

net 43.43.23.3 0.0.0.0 a 0
```

라우터/스위치 라우팅 설정 (2)

R4

```
router os 1
router-id 43.43.0.4

net 43.43.24.4 0.0.0.0 a 0
```

R5

```
router os 1
router-id 43.43.0.5
net 43.43.25.5 0.0.0.0 a 0
redistribute ei 43 subnets

router ei 43
no auto-summary
net 43.43.55.5 0.0.0.0
net 43.43.155.5 0.0.0.0
redistribute os 1 met 1 1 1 1 1
```

라우터/스위치 라우팅 설정 (3)

SW1

```
router ei 43
no auto-summary
net 43.43.155.250 0.0.0.0
```

SW2

```
router ei 43
no auto-summary
net 43.43.55.250 0.0.0.0
```

SW4

```
ip route 0.0.0.0 0.0.0.0
43.43.2.253
```

라우팅 설정 확인 R2(라우팅 테이블)

```
R2(config)#do sh ip ro
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 43.43.5.253 to network 0.0.0.0

     222.222.222.0/32 is subnetted, 1 subnets
C       222.222.222.222 is directly connected, Loopback100
     43.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       43.43.0.2/32 is directly connected, Loopback0
S       43.43.3.0/24 [1/0] via 43.43.5.253
C       43.43.5.0/24 is directly connected, FastEthernet0/0
C       43.43.23.0/24 is directly connected, Serial0/1
C       43.43.24.0/24 is directly connected, Serial0/0
C       43.43.25.0/24 is directly connected, FastEthernet0/1
O E2    43.43.55.0/24 [110/20] via 43.43.25.5, 00:02:11, FastEthernet0/1
O E2    43.43.155.0/24 [110/20] via 43.43.25.5, 00:02:11, FastEthernet0/1
S*   0.0.0.0/0 [1/0] via 43.43.5.253
```

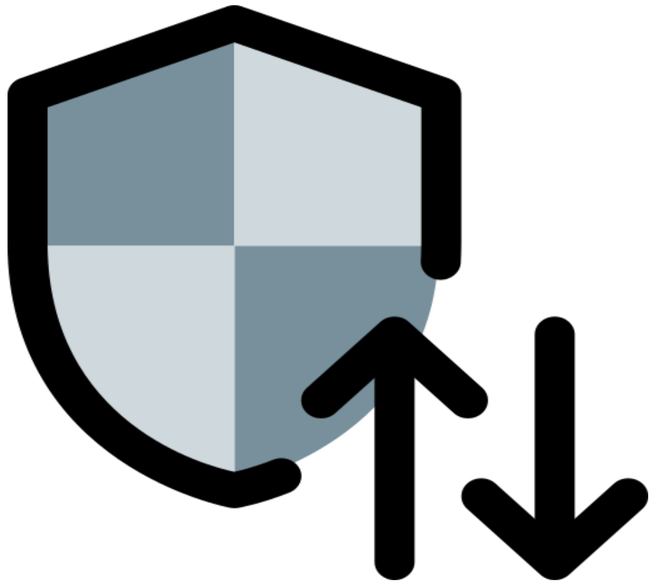
Chapter 4

방화벽 설정 및 이중화

Chapter 4-(1)

이중화의 의미

장애에 대비하기 위해 두 개의 동일한 장비를 사용하는 것으로 이중화가 구성된 방화벽은 대상 인터페이스들의 상태를 모니터링하다가 장애가 발생하는 경우 failover(역할 교대)가 동작한다.



이중화 종류(A-S, A-A)

Active-Standby 이중화

한 장비에서만 트래픽을 처리하며 단일 및 다중 컨텍스트를 모두 설정할 수 있다.

Active-Active 이중화

두 장비 모두 트래픽을 처리하는 부하분산이 가능하며 다중 컨텍스트만 설정할 수 있다.

Chapter 4-(3)

방화벽 설정 및 이중화-기본 설정 FW1

FW1

```
activation-key 0x4a3ec071 0x0d86fbf6
0x7cb1bc48 0x8b48b8b0 0xf317c0b5

reload

mode multiple

int g0
no sh

int g1
no sh

int g2
no sh

int g3
no sh

no failover

failover lan unit primary
failover lan interface fover g3
failover link fover g3

failover interface ip fover 43.43.100.100 255.255.255.0 standby
43.43.100.101

failover
```

방화벽 설정 및 이중화-기본 설정 FW2

FW2

```
activation-key 0x4a3ec071 0x0d86fbf6
0x7cb1bc48 0x8b48b8b0 0xf317c0b5

reload

mode multiple

int g0
no sh

int g1
no sh

int g2
no sh

int g3
no sh

no failover

failover lan unit secondary
failover lan interface fover g3
failover link fover g3

failover interface ip fover 43.43.100.100 255.255.255.0 standby
43.43.100.101

failover
```

방화벽 설정 및 이중화-기본 설정 확인

FW1

```
FW1(config)# sh failover
Failover On
Failover unit Primary
Failover LAN Interface: fover GigabitEthernet3 (up)
Unit Poll frequency 1 seconds, holdtime 15 seconds
Interface Poll frequency 5 seconds, holdtime 25 seconds
Interface Policy 1
Monitored Interfaces 0 of 60 maximum
Version: Ours 8.4(2), Mate 8.4(2)
Last Failover at: 04:28:13 UTC Jul 16 2025
        This host: Primary - Active
                Active time: 84 (sec)
        Other host: Secondary - Standby Ready
                Active time: 0 (sec)

Stateful Failover Logical Update Statistics
        Link : fover GigabitEthernet3 (up)
        Stateful Obj    xmit       xerr       rcv        rerr
        General         11         0          10         0
        sys cmd         10         0          10         0
        up time         0          0          0          0
        RPC services    0          0          0          0
        TCP conn        0          0          0          0
        UDP conn        0          0          0          0
        ARP tbl         0          0          0          0
        Xlate_Timeout   0          0          0          0
<--- More --->
```

FW2

```
FW2(config)# sh failover
Failover On
Failover unit Secondary
Failover LAN Interface: fover GigabitEthernet3 (up)
Unit Poll frequency 1 seconds, holdtime 15 seconds
Interface Poll frequency 5 seconds, holdtime 25 seconds
Interface Policy 1
Monitored Interfaces 0 of 60 maximum
Version: Ours 8.4(2), Mate 8.4(2)
Last Failover at: 04:24:24 UTC Jul 16 2025
        This host: Secondary - Standby Ready
                Active time: 0 (sec)
        Other host: Primary - Active
                Active time: 96 (sec)

Stateful Failover Logical Update Statistics
        Link : fover GigabitEthernet3 (up)
        Stateful Obj    xmit       xerr       rcv        rerr
        General         12         0          13         0
        sys cmd         12         0          12         0
        up time         0          0          0          0
        RPC services    0          0          0          0
        TCP conn        0          0          0          0
        UDP conn        0          0          0          0
        ARP tbl         0          0          0          0
        Xlate_Timeout   0          0          0          0
<--- More --->
```

방화벽 설정 및 이중화-컨텍스트 생성 및 할당 FW1

FW1

```
context c1
config-url c1.cfg
allocate-interface g2
allocate-interface g0
allocate-interface g1

context c2
config-u c2.cfg
allocate-interface g2
allocate-interface g0
```

```
FW1(config-ctx)# sh context
Context Name      Class      Interfaces          URL
*admin            default                        disk0:/admin.cfg
 c1               default    GigabitEthernet0,   disk0:/c1.cfg
                             GigabitEthernet1,
                             GigabitEthernet2
 c2               default    GigabitEthernet0,   disk0:/c2.cfg
                             GigabitEthernet2

Total active Security Contexts: 3
```

Chapter 4-(7) FW1 context c1 인터페이스 설정

방화벽 설정 및 이중화-컨텍스트 주소 설정 FW1

FW1

```
changeto context c1

int g2
nameif inside
ip add 43.43.2.253
255.255.255.0 standby
43.43.2.254

int g1
nameif DMZ3
security-level 100
ip add 43.43.3.253
255.255.255.0 standby
43.43.3.254

int g0
nameif outside
security-level 0
ip add 43.43.5.253 255.255.255.0 standby 43.43.5.254

route outside 0 0 43.43.5.2
route inside 150.1.43.0 255.255.255.0 43.43.2.250
route DMZ 43.43.0.1 255.255.255.255 43.43.3.1
```

방화벽 설정 및 이중화-Access List 설정 및 확인 FW1

FW1

changeto context c1

access-l acl_oi per icmp a a
access-g acl_oi in int outside

same-security-traffic per inter-interface

```
FW1/c1# sh access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096)
            alert-interval 300
access-list acl_oi; 1 elements; name hash: 0x4bf52f3b
access-list acl_oi line 1 extended permit icmp any any (hitcnt=0) 0x865e8c90
FW1/c1#
```

Chapter 4-(9)

방화벽 설정 및 이중화-Active&Active 모드 설정 FW1

FW1

```
changeto sys
no failover

failover group 1
preempt

failover group 2
secondary

preempt
```

```
context c2
join-failover-group 1

context c1
join-failover-group 2

failover
failover active
```

방화벽 설정 및 이중화-Active&Active 모드 설정 확인

FW1

```
FW1(config)# sh failover
Failover On
Failover unit Primary
Failover LAN Interface: fover GigabitEthernet3 (up)
Unit Poll frequency 1 seconds, holdtime 15 seconds
Interface Poll frequency 5 seconds, holdtime 25 seconds
Interface Policy 1
Monitored Interfaces 3 of 60 maximum
Version: Ours 8.4(2), Mate 8.4(2)
Group 1 last failover at: 04:41:13 UTC Jul 16 2025
Group 2 last failover at: 04:41:40 UTC Jul 16 2025

  This host:    Primary
  Group 1       State:          Active
                Active time:    39 (sec)
  Group 2       State:          Active
                Active time:    12 (sec)

                  cl Interface inside (43.43.2.253): Normal (Waiting)
                  cl Interface DMZ3 (43.43.3.253): Normal (Waiting)
                  cl Interface outside (43.43.5.253): Normal (Waiting)

  Other host:   Secondary
  Group 1       State:          Standby Ready
                Active time:    0 (sec)
  Group 2       State:          Failed
                Active time:    31 (sec)

                  cl Interface inside (43.43.2.254): Normal (Monitored)
                  cl Interface DMZ3 (43.43.3.254): Failed (Waiting)
                  cl Interface outside (43.43.5.254): Normal (Monitored)
```

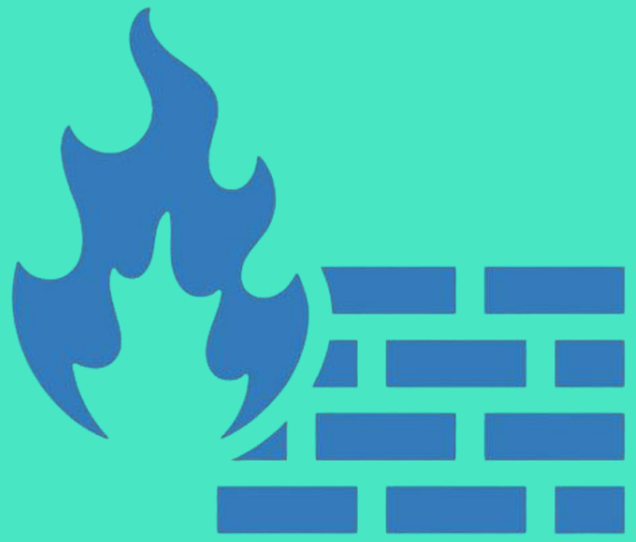
FW2

```
FW2(config)# sh failover
Failover On
Failover unit Secondary
Failover LAN Interface: fover GigabitEthernet3 (up)
Unit Poll frequency 1 seconds, holdtime 15 seconds
Interface Poll frequency 5 seconds, holdtime 25 seconds
Interface Policy 1
Monitored Interfaces 3 of 60 maximum
Version: Ours 8.4(2), Mate 8.4(2)
Group 1 last failover at: 04:41:14 UTC Jul 16 2025
Group 2 last failover at: 04:41:40 UTC Jul 16 2025

  This host:    Secondary
  Group 1       State:          Standby Ready
                Active time:    0 (sec)
  Group 2       State:          Failed
                Active time:    31 (sec)

                  cl Interface inside (43.43.2.254): Normal (Monitored)
                  cl Interface outside (43.43.5.254): Normal (Monitored)
                  cl Interface DMZ3 (43.43.3.254): Failed (Waiting)

  Other host:   Primary
  Group 1       State:          Active
                Active time:    188 (sec)
  Group 2       State:          Active
                Active time:    161 (sec)

                  cl Interface inside (43.43.2.253): Normal (Monitored)
                  cl Interface outside (43.43.5.253): Normal (Monitored)
                  cl Interface DMZ3 (43.43.3.253): Normal (Waiting)
```

Thank you for Watching